Eg :-

Test data

	Actual label	Predicted Score	Predicted label
1	1	0.85	1
2	0	0.6	1
3	1	0.7	1
4	1	0.4	0
5	0	0.55	1
6	1	0.5	1
7	0	0.65	1
8	0	0.35	0
9	1	0.6	1
10	0	0.2	0

Threshold = 0.5

Compute :-

1) Confusion matrix
2) Accuracy
3) Precision
4) Recall / Sensitivity
5) F_1 - score
6) ROC curve

(0.7, 0.5, 0.4, 0.2)

TP 4	FP 3
FN 1	TN 2

$$\text{Accuracy} = \frac{6}{10} = 0.6$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{4}{7} = 0.57$$

$$\text{Recall / Sensitivity} = \frac{TP}{TP+FN} = \frac{4}{5} = 0.8$$

$$F_1 = \frac{2 \times 0.57 \times 0.8}{0.57 + 0.8} = \frac{0.912}{1.37} = 0.665$$

$$\text{specificity} = \frac{2}{5} \Rightarrow \frac{TN}{TN+FP}$$

ROC :-

Threshold	specificity	sensitivity	1 - specificity
0.7	1	0.4	0
0.5	0.4	0.8	0.6
0.4	0.4	1	0.6
0.2	0	1	1

	Actual label	for 0.7 Threshold Predicted label	for 0.2	for 0.4
1	1	1 ✓	1 ✓	1 ✓
2	0	0 ✓	1 ✗	1
3	1	1 ✓	1 ✓	1 ✓
4	1	0 ✗	1 ✓	1 ✓
5	0	0 ✓	1 ✗	1
6	1	0 ✗	1 ✓	1 ✓
7	0	0 ✓	1 ✗	1
8	0	0 ✓	1 ✗	0
9	1	0 ✗	1 ✓	1 ✓
10	0	0 ✓	1 ✗	0

TP	FP
2	0
FN	TN
3	5

Specificity $= \frac{5}{5} = 1$

sensitivity $= \frac{2}{5} = 0.4$

1-specificity $= 0$

TP	FP
5	5
FN	TN
0	0

specificity $= 0$

Sensitivity $= \frac{5}{5} = 1$

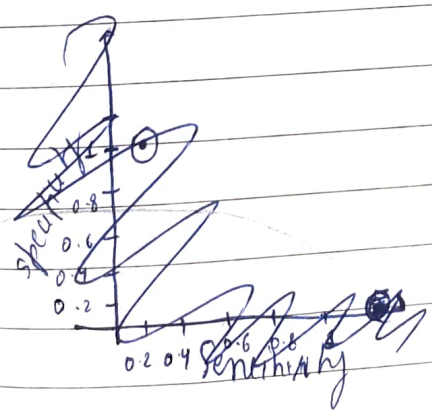
1-specificity $= 1$

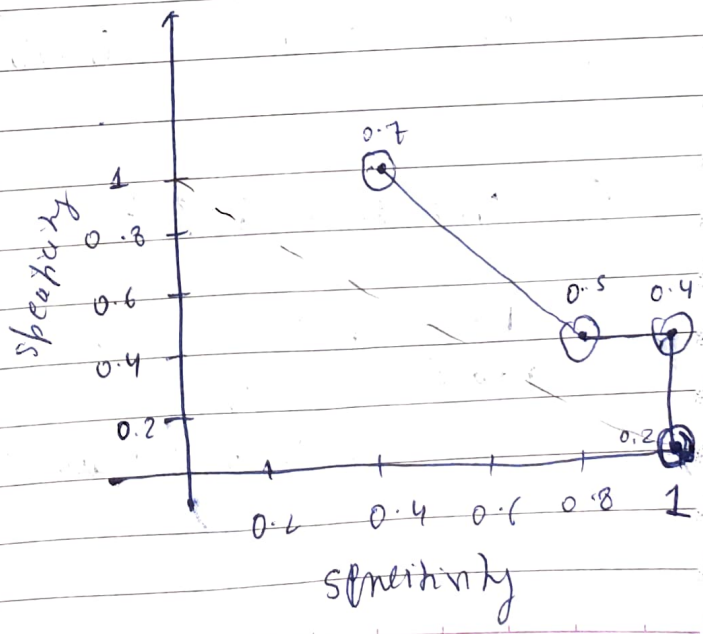
TP	FP
5	3
FN	TN
0	2

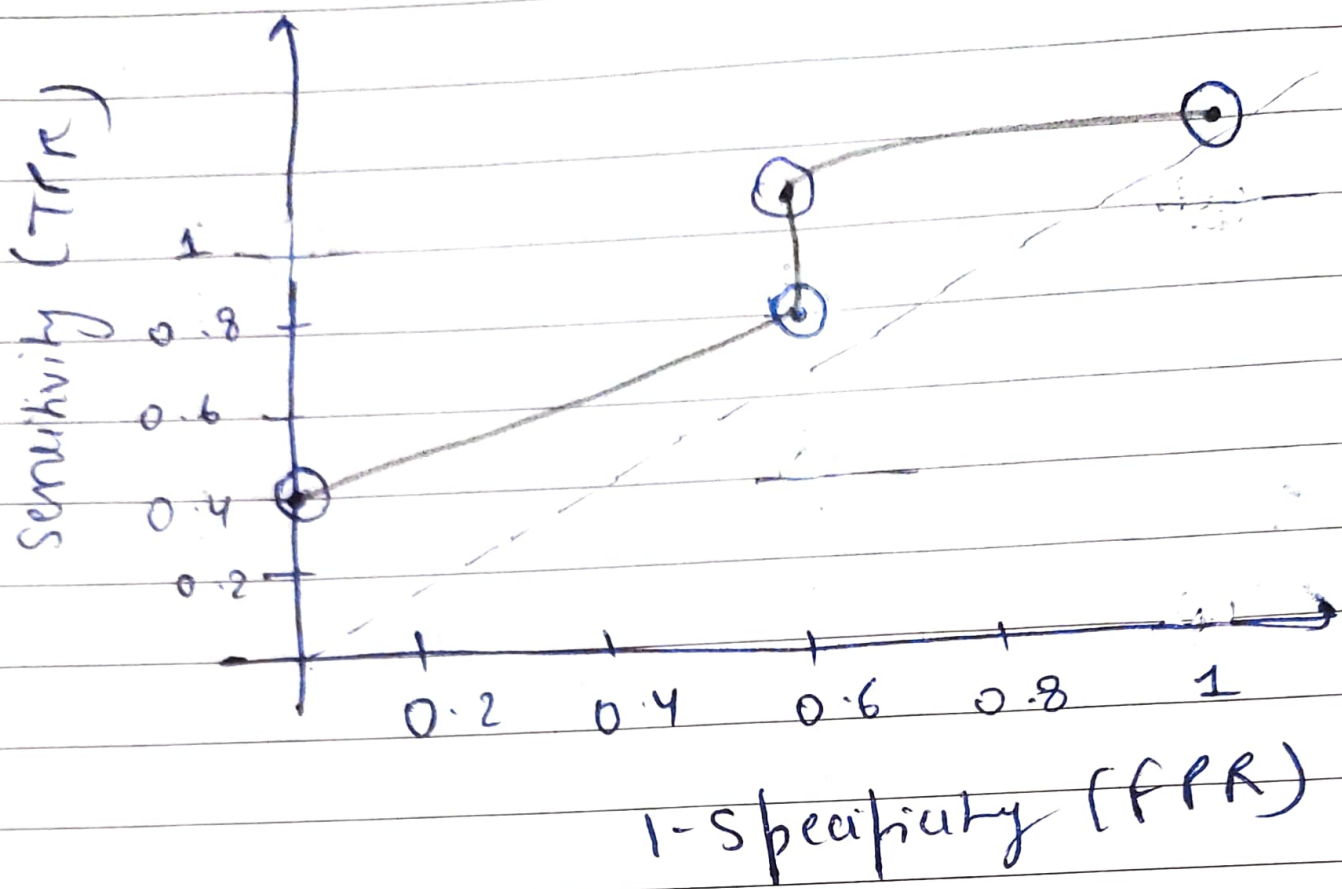
Specificity $= \frac{2}{5} = 0.4$

Sensitivity $= \frac{5}{5} = 1$

1-specificity $= 0.6$






Sensitivity (TPR)
1
0.8
0.6
0.4
0.2
0.2
0.4
0.6
0.8
1
1-Specificity (FPR)